

foreach Loop, break, continue, Nested Loop, Array

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foreach Loop

```
foreach ($array as $key=>$value) {  
    code to be executed;  
}
```

```
foreach ($array as $value) {  
    code to be executed;  
}
```

Example

Example - array:

```
$arr = ["usa","australia","canada"];  
foreach($arr as $val) {  
    echo $val;  
}
```

Example - associative array:

```
$arr = [ "username" => "arefin", "age" => 30, "password" => "1234" ];  
foreach($arr as $key=>$val) {  
    echo "KEY is: {$key} and Value is: {$val}";  
}
```

break

- The break statement is used to jump out of a loop.
- break;

Example:

```
for($i=1;$i<=10;$i++) {  
    if($i==5) {  
        break;  
    }  
    echo $i;  
}
```

continue

- The continue statement breaks one iteration (in the loop), if a specified condition occurs, and continues with the next iteration in the loop.
- continue;

Example:

```
for($i=1;$i<=10;$i++) {  
    if($i==5) {  
        continue;  
    }  
    echo $i;  
}
```

Nested Loop

- A nested loop is a loop that is inside of another loop.
- This allows the inner loop to execute a certain number of times for each iteration of the outer loop.

Example:

1. Pyramid
2. Multiplication Table

Example - Multiplication Table

```
for($i=1;$i<=10;$i++) {  
    echo "i=".$i." : ";  
    for($j=1;$j<=10;$j++) {  
        echo $j.' ';  
    }  
    echo "\n";  
}
```

Output:

```
i=1: 1 2 3 4 5 6 7 8 9 10  
i=2: 1 2 3 4 5 6 7 8 9 10  
i=3: 1 2 3 4 5 6 7 8 9 10  
i=4: 1 2 3 4 5 6 7 8 9 10  
i=5: 1 2 3 4 5 6 7 8 9 10  
i=6: 1 2 3 4 5 6 7 8 9 10  
i=7: 1 2 3 4 5 6 7 8 9 10  
i=8: 1 2 3 4 5 6 7 8 9 10  
i=9: 1 2 3 4 5 6 7 8 9 10  
i=10: 1 2 3 4 5 6 7 8 9 10
```

Example - Pyramid 1

```
for($i=1;$i<=4;$i++) {  
    for($j=1;$j<=$i;$j++) {  
        echo $j." ";  
    }  
    echo "\n";  
}
```

Output:

```
1  
1 2  
1 2 3  
1 2 3 4
```


Example - Pyramid 2

```
for($i=1;$i<=4;$i++) {  
    for($j=1;$j<=$i;$j++) {  
        echo $i." ";  
    }  
    echo "\n";  
}
```

Output:

```
1  
2 2  
3 3 3  
4 4 4 4
```

Example - Pyramid 3

```
for($i=1;$i<=4;$i++) {  
    for($j=$i;$j>=1;$j--) {  
        echo $j." ";  
    }  
    echo "\n";  
}
```

Output:

```
1  
2 1  
3 2 1  
4 3 2 1
```

Example - Pyramid 4

```
$total_rows = 5;
for($i=1;$i<=$total_rows;$i++) {
    for($k=$total_rows-1;$k>=$i;$k--) {
        echo " ";
    }
    for($j=$i;$j>=1;$j--) {
        echo $j." ";
    }
    for($m=2;$m<=$i;$m++) {
        echo $m." ";
    }
    echo "\n";
}
```

Output:

```
    1
  2 1 2
3 2 1 2 3
4 3 2 1 2 3 4
5 4 3 2 1 2 3 4 5
```

Array

- An array stores multiple values in one single variable.

2 Formats:

```
$arr = array(1,2,3);
```

```
$arr = [1,2,3]
```

Types of Array:

1. Indexed Array - Arrays with a numeric index
2. Associative Array - Arrays with named keys
3. Multidimensional Array - Arrays containing one or more arrays

Indexed Array

Example - numeric:

```
$arr = [12, 44, 55, 19, 34, 89];  
echo $arr[0]; // 12  
echo $arr[1]; // 44
```

Example - string:

```
$cities = ["Khulna", "Dhaka", "Rajshahi"];  
echo $cities[0]; // Khulna  
echo $cities[1]; // Dhaka
```

Example - numeric elements - for loop:

```
$arr = [12, 44, 55, 19, 34, 89];  
for($i=0;$i<count($arr);$i++) {  
    echo $arr[$i];  
}
```

Indexed Array (Continue ...)

Example - numeric elements - foreach loop:

```
$arr = [12, 44, 55, 19, 34, 89];  
foreach($arr as $val) {  
    echo $val;  
}
```

Example - string elements - for loop:

```
$cities = ["Khulna", "Dhaka", "Rajshahi"];  
for($i=0;$i<count($cities);$i++) {  
    echo $cities[$i];  
}
```

Example - string elements - foreach loop:

```
$cities = ["Khulna", "Dhaka", "Rajshahi"];  
foreach($cities as $val) {  
    echo $val;  
}
```